unit 3. Liked list.

- Title : movie watchlist manager using singly linked list

- problem statement :

Develop a movie watchlist manager using a singly linked list to store movies that a user plans to watch. The system should allow the user to add, remove and display movies in the watchlist.

- Objectives :

1. To understand the concept of singly linked list
2. To create and manage movie nodes dynamically.
3. To add movies to the watchlist.
4. To remove movies from the watchlist.
5. To Display all movies in the watchlist.
6. To understand pointers and dynamic memory allocation.

- Algorithm :

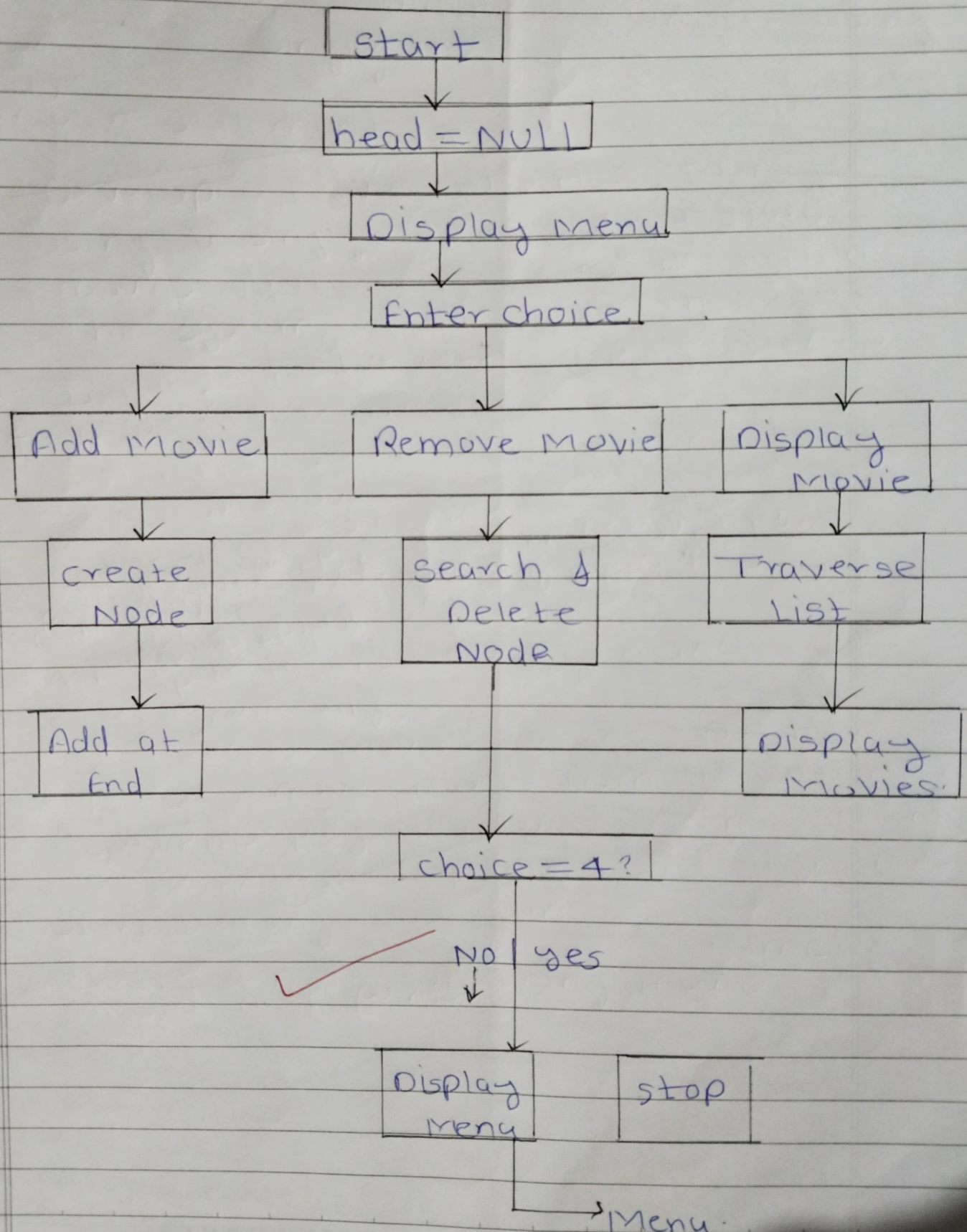
1. Start
2. Create a node containing movie name and next pointer.

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3. Initialize head = NULL.
4. Display the menu:
   - Add Movie:
   - Remove Movie:
   - Display Movies:
   - Exit
5. for Add Movie, create a new node and add it at the end of the linked list.
6. For Remove Movie, search for the movie and delete its node.
7. For Display Movies, traverse the linked list and display all movie names.
8. Repeat the operations until the user selects Exit.
9. stop.

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- Flowchart:



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```
#include <iostream>
#include <string>
using namespace std;

// Node structure
struct Node {
    string movie;
    Node* next;
};

// Add movie at the end
void addMovie(Node*& head, string movie) {
    Node* newNode = new Node();
    newNode->movie = movie;
    newNode->next = NULL;

    if (head == NULL) {
        head = newNode;
    }
    else {
        Node* temp = head;

        while (temp->next != NULL) {
            temp = temp->next;
        }

        temp->next = newNode;
    }
```

```
    cout << "Movie added successfully!\n";
}

// Remove movie
void removeMovie(Node*& head, string movie) {
    if (head == NULL) {
        cout << "Watchlist is empty.\n";
        return;
    }

    // If first movie is to be removed
    if (head->movie == movie) {
        Node* temp = head;
        head = head->next;
        delete temp;

        cout << "Movie removed successfully!\n";
        return;
    }

    Node* temp = head;

    while (temp->next != NULL && temp->next->movie != movie) {
        temp = temp->next;
    }

    if (temp->next == NULL) {
```

```
    cout << "Movie not found.\n";
  }
  else {
    Node* deleteNode = temp->next;
    temp->next = deleteNode->next;
    delete deleteNode;

    cout << "Movie removed successfully!\n";
  }
}

// Display watchlist
void displayMovies(Node* head) {
  if (head == NULL) {
    cout << "Watchlist is empty.\n";
    return;
  }

  cout << "\n--- Movie Watchlist ---\n";

  Node* temp = head;
  int count = 1;

  while (temp != NULL) {
    cout << count << ". " << temp->movie << endl;
    temp = temp->next;
    count++;
  }
```

```
}

int main() {
    Node* head = NULL;
    int choice;
    string movie;

    do {
        cout << "\n===== Movie Watchlist Manager =====\n";
        cout << "1. Add Movie\n";
        cout << "2. Remove Movie\n";
        cout << "3. Display Watchlist\n";
        cout << "4. Exit\n";
        cout << "Enter your choice: ";
        cin >> choice;

        switch (choice) {
            case 1:
                cout << "Enter movie name: ";
                cin.ignore();
                getline(cin, movie);
                addMovie(head, movie);
                break;

            case 2:
                cout << "Enter movie name to remove: ";
                cin.ignore();
                getline(cin, movie);
```

```
            removeMovie(head, movie);
            break;

        case 3:
            displayMovies(head);
            break;

        case 4:
            cout << "Exiting program...\n";
            break;

        default:
            cout << "Invalid choice!\n";
    }

} while (choice != 4);

return 0;
}
```


File Edit Selection View Go Run Terminal Help
c programing code
Problems Output Debug Console Terminal Ports
Enter choice: 1
Enter movie name: avengers
Movie added successfully.
--- Movie Watchlist Manager ---
1. Add Movie
2. Remove Movie
3. Display Movies
4. Exit
Enter choice: 1
Enter movie name: intersteller
Movie added successfully.
--- Movie Watchlist Manager ---
1. Add Movie
2. Remove Movie
3. Display Movies
4. Exit
Enter choice: 3
Movie Watchlist:
avengers <-> intersteller <-> NULL
--- Movie Watchlist Manager ---
1. Add Movie
2. Remove Movie
3. Display Movies
4. Exit
Enter choice: 4
Program ended.
PS D:\c programing code>
Chat
powersh...
Sessi...
powershell
powershell
powershell
Tip: Use /create-agent to scaffold a custom agent for your workflow.
02.MovieWatchli
1dowhile.c
Describe what to b...
Local
BLACKBOX Agent
Open Website
Ln 144, Col 2
Spaces: 4
UTF-8
CRLF
C++
Go Live
AI Code Chat
Win32
Search
ENG
IN
63%
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